Deep Armocromia: A Novel Dataset for Face Seasonal Color Analysis and Classification

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Abstract. Recent advancements in Artificial Intelligence and Computer Vision, in particular Deep Learning (DL), have transformed the analysis of human faces, enabling different tasks, ranging from classification to synthesis. Despite these advancements, color analysis in face images remains underexplored, especially concerning well-defined datasets and frameworks tailored to specific methodologies such as Season Color Analysis or Armocromia. Armocromia combines qualitative and quantitative approaches to determine personal color palettes based on an individual's skin, hair, and eye color; for this, it is vastly adopted in the fashion world. However, we found a lack of datasets to train DL models to automatically discriminate among these classes. To this date, we introduce Deep Armocromia, a novel dataset comprising labeled face images categorized according to Armocromia Flow Theory, with a strict annotation protocol. We conduct experiments to validate the effectiveness of DL models in discriminating among Armocromia classes optimized on Deep Armocromia. Results underscore the challenges inherent to Armocromia classification and highlight opportunities for advancing DL architectures and optimization methodologies.

Keywords: Face Analysis · Deep Learning · Armocromia · Dataset

1 Introduction

In recent years, artificial intelligence and computer vision approaches have revolutionized the way we extract and analyze patterns within images containing human faces, with different objectives [3, 7, 11, 15, 26, 27]. Considering discriminative face analysis, the Face Recognition task is one of the most explored and amounts to a biometric technique for identity authentication [27]. On a similar line, Face Re-Identification aims at retrieving a person of interest across multiple non-overlapping cameras [30]. Moving to generative approaches, paradigms such as GANs and Diffusion Models have been employed to learn and generate complex face image distributions, providing fine-grained control over Face Generation [9].

To achieve this variety of tasks a fundamental step amounts to feature extraction, which aims at providing robust features representing visual cues [27]. Such features are nowadays learned and extracted by parametric Deep Learning (DL) architectures, such as Convolutional Neural Networks (CNN) and Vision Transformers (ViTs), which are optimized on one of the aforementioned tasks or using self-supervised approaches [18, 27]. Those features learned implicitly, often contain representation related to color, employed for different aims, like emotional state detection, outdoor season discrimination, or injecting color constancy [4,5,14,20,28]. However, we found a lack of frameworks, training pipelines, and datasets to optimize DL models to discriminate a peculiar set of classes within a recently emerged face color analysis task: Armocromia.

Armocromia [2], or seasonal color analysis, is a mixed qualitative-quantitative method used to determine the most compatible colors for an individual based on their natural complexion, eye, and hair color, introduced by the Italian Image Institute and Rossella Migliaccio [17]. Such color palettes are then exploited to enhance a person's fashion style by aligning it with their unique chromatic characteristics [16, 25].

The Armocromia analysis process is mainly composed of two steps: (i) **Personal color properties**, to determine the skin's, hairs' and eyes' undertone, and lightness and evaluate the contrast between skin, hair, and eyes, and measure the clarity of the person's natural coloring; (ii) **Classification into Seasonal Types and Subtypes**, which analyzes the extracted color properties to classify them into one of the four main seasons (i.e., Winter, Spring, Summer, Autumn) and possible sub-types (i.e., deep, bright, light, soft, warm, cool) [16]. Considering such a process, we hypothesize that such features and discriminative properties could be learned implicitly by DL models. However, there is a lack of datasets designed to verify such a possibility, following strict and robust Armocromia protocols.

To this date, we here introduce Deep Armocromia [3], a novel dataset composed of Face images, labeled according to Armocromia rules in the four main Seasons and their Sub-Types. Such dataset was then employed in an experimental campaign to compare and validate the usage of pre-trained DL models to discriminate among Armocromia classes, following a classic fine-tuning approach [8]. Results showed how Armocromia classification amounts to a challenging task, motivating further advancements and implementations for DL feature extractors and classifiers.

To summarize, the contributions of this work amount to:

- **Introduction of Deep Armocromia dataset**: A novel dataset of face images labeled according to Armocromia rules in main seasons and sub-types, facilitating the training and evaluation of deep learning models tailored to Armocromia protocols.
- **Experimental validation of DL models**: Comparison and validation of pre-trained DL models' effectiveness in discriminating among Armocromia

[3] The dataset is available at `https://github.com/lorenzo-stacchio/Deep-Armocromia`

classes, demonstrating their potential for learning implicit features and discriminative properties.

2 Related Works

In this Section, we review the most adopted Face Analysis datasets and the works that analyzed color properties within Face images.

2.1 Face Datasets

CelebA The CelebFaces Attributes Dataset is a large-scale face attributes dataset consisting of over 200,000 celebrity images, each annotated with 40 attribute labels [11]. These images exhibit significant variations in pose and background clutter, enhancing the dataset's robustness for diverse applications. This dataset was adopted for a variety of computer vision tasks, including face attribute recognition, face recognition, face detection, facial landmark localization, and face synthesis. The extensive diversity, quantity, and rich annotations of CelebA make it an invaluable resource for training and testing models in these domains.

LAION-Face Introduced in [31] is a specialized subset of the LAION-400M [22], focusing on images containing human faces. It comprises 50 million image-text pairs, referred to as LAION-Face 50M, curated through face detection techniques to identify images featuring human faces. Additionally, a streamlined subset, LAION-Face 20M, is provided for efficient evaluation, containing 20 million image-text pairs. This dataset facilitates research in face recognition, expression analysis, and other applications in computer vision by offering a large-scale, diverse collection of annotated image-text pairs.

DigiFace-1M This dataset is a comprehensive collection of over one million synthetic face images, specifically designed to enhance face recognition research [1]. This dataset includes 720,000 images encompassing 10,000 identities, with each identity having 72 images rendered across four different sets of accessories. The diverse accessory sets and substantial volume of images make DigiFace-1M a valuable resource for developing and evaluating robust facial recognition systems.

Differently from all the mentioned datasets that were designed to train and optimize DL models to analyze different face-related tasks, to the best of our knowledge, we here introduced the first dataset exclusively designed for Armocromia classification.

2.2 Deep Learning Face Color Analysis

Different related works have analyzed face images focusing on color features [19, 23, 25, 32]. Authors of [23] explore micro emotions classification with a novel approach that detects and classifies micro emotions based solely on facial color

changes due to blood flow, demonstrating their efficacy compared to motion-based methods. For example, [19] analyzed the effectiveness of geometric and color features, individually and in combination, for face recognition using machine learning approaches. Color features were derived from normalized RGB values, mean RGB values, and RGB variance that were demonstrated to enhance face recognition systems on a novel-defined dataset. Instead, [32] investigates automatic classification of facial skin color using hyperspectral imaging due to the complexity of the Pantone scale's high granularity labels, encompassing up to 110 facial skin-color types, demonstrating that their method identifies skin chromaticity and brightness as distinct features that were used for the final color classification, achieving state of the art result.

Despite these relevant contributions, we found a lack of works focusing on exploiting face color features in the context of Seasonal Color Analysis. To the best of our knowledge, the only work that analyzed this context amounts to [25]. In this work, authors tackled the issue of e-commerce clothing recommendations by developing an innovative algorithm that integrates user-specific characteristics, including face color, pupil color, and face shape, classifying those with the established 4-Season Color System of Armocromia, that are then used for clothing suggestions. To train their classifiers, the authors crawled a collection of 1000 images from the web and cataloged that into the 4-Season Color System. However, this dataset was not released and there is no information about how it was constructed or its distribution. Moreover, this dataset does not take into account the Flow Theory and a strict cataloging protocol. Unlike previous studies, we have assembled a novel dataset consisting of images of public figures. This dataset is uniquely annotated, including not only the classic 4-Season Color classifications but also their respective Sub-Types, and is freely available for use.

3 Dataset

The Deep Armocromia dataset has been collected since March 2024 to study how discriminative visual cues can be learned automatically by Machine and Deep Learning models at the University of Macerata [4]. This produced a digitized collection of Face images comprising ca. 5000 photos labeled according to Armocromia rules. In particular, those pictures were labeled by annotators under the supervision of an Armocromia expert. Such a dataset is, to the best of our knowledge, the first dataset for Armocromia analysis based on Facial Features. To better understand the domain knowledge that is necessary to collect such a dataset, we will first review in detail what is Armocromia and its classifications.

3.1 Armocromia Domain Knowledge

Armocromia is an interdisciplinary field derived from the color theory that meticulously examines an individual's skin, eye, and hair colors to determine their ideal

[4] `https://www.unimc.it`

color palette [10, 16]. Armocromia's most interesting aspect lies in its ability to work in close association with fashion while simultaneously offering an alternative perspective. Fashion trends, driven by major brands and influencers, often promote styles and colors that may not suit everyone, leading to a homogenized appearance. Armocromia counters this by focusing on the individual, allowing for a more personalized approach to fashion.

Armocromia Color Analysis and Seasons The objective of Armocromia is to identify and enhance individual colors by leveraging principles of color harmony. This method involves recognizing an individual's color characteristics to identify specific traits [16]. The primary color characteristics of interest are fourfold, derived from the interplay of skin, eyes, and hair [16]. These include undertone or temperature, which can be warm or cool; value or tone, which ranges from light to dark; contrast, which refers to the degree of contrast among the skin, eyes, and hair; and intensity or brightness of colors, which can be strong or subdued. Analyzing and combining these characteristics allows for the classification of individuals into four macro-groups, corresponding to different color types known as "seasons", which are differentiated into "Spring", "Summer", "Autumn", and "Winter".

Armocromia Flow Theory and Season Sub-Types The Flow Theory expands the traditional seasonal color analysis by introducing subgroups, allowing for a more nuanced representation of chromatic diversity [16]. This theory posits that individuals can exhibit characteristics of two different seasons, leading to a "sliding" between seasons. Each season is divided into three subgroups, creating twelve possible categories, which better capture the range of natural color variations. This sliding mechanism acknowledges predominant traits that do not fit neatly into a single category [16]. According to the Flow Theory, after establishing the primary season, the subgroup is determined based on the most prominent characteristic of the individual, according to Temperature, Contrast, and Intensity. Table 1 provides a recap of the Season and the possible value that it could be paired with, for a total of twelve combinations (e.g., Spring-Warm, Summer-Light, Autumn-Soft, and Winter-Bright).

Season	**Temperature**	**Contrast**	**Intensity**
Spring	Warm	Light	Bright
Summer	Cool	Light	Soft
Autumn	Warm	Deep	soft
Winter	Cool	Deep	Bright

Table 1: Flow Theory classification system, organized by Season and Sub-Types.

Since this classification is the most flexible and adopted within the community, we here state that all our samples included in Deep Armocromia were classified according to it.

3.2 Image Collection and Annotation Process

The annotation process followed (and keeps following) a simple but strict protocol involving the following steps: (a) Armocromia rules study; (b) Visual Cluster collection; (c) Image Annotation.

Armocromia rules study During the first lecture, the Armocromia classification process and its definition were furnished, exploring different examples coming from well-established experts. In particular, concepts like undertone, value, contrast, temperature, and intensity of colors and human faces were covered. During a second lecture, the Armocromia annotation process was detailed, furnishing the students with a strict and precise protocol for labeling. Such protocol visually analyzes and matches an unlabeled Face by analyzing hair, skin, and eye visual properties to already defined ground truth for all the considered Armocromia classes. Such ground truth was first collected by analyzing the case studies coming from Armocromia experts, like Rossella Migliaccio [5].

Armocromia Visual cluster Collection Starting from the knowledge acquired during the case study analysis, annotators first acquire Images of known celebrities like actors, artists, singers, influencers, and models that already have a well-defined Face Armocromia analysis. The pictures were collected mainly from social media. Such a process resulted in the collection of pictures from more than 250 identities, resulting in the collection of approximately 1900 images (named as Pivotal Armocromia Set). Each identity was then classified according to one of the Season classes and its respective sub-type.

Image Annotation Each of the collected identities was used as visual pivots to annotate a subset of one of the most used Faces datasets, which also has the nearest data distribution to our already collected images: the CelebA dataset. In practice, we randomly sampled 3000 images from the CelebA (to avoid as much as possible skin, eyes, and in general facial features bias) that were then annotated by our annotators, exploiting the learned Armocromia rules and already collected and annotated identities. This second collection and annotation step has converged into approximately 5000 annotated pictures.

We here highlight that all these described processes show the uniqueness of these collected datasets: since only trained annotators could have the ground truth, it is not possible to resort to just any standard labeling services (e.g., Amazon SageMaker [6]).

[5] `https://rossellamigliaccio.com/`
[6] `https://aws.amazon.com/pm/sagemaker/?nc1=h_ls`

3.3 Data Cleaning and Processing

We first identified duplicate images by comparing their perceptual hashed representation calculated with the Average Perceptual hash algorithm with a size of 16 bits [6]. Then, considering that Armocromia requires analysis only of face-related features, we implement a data processing step to isolate only those pixels related to it.

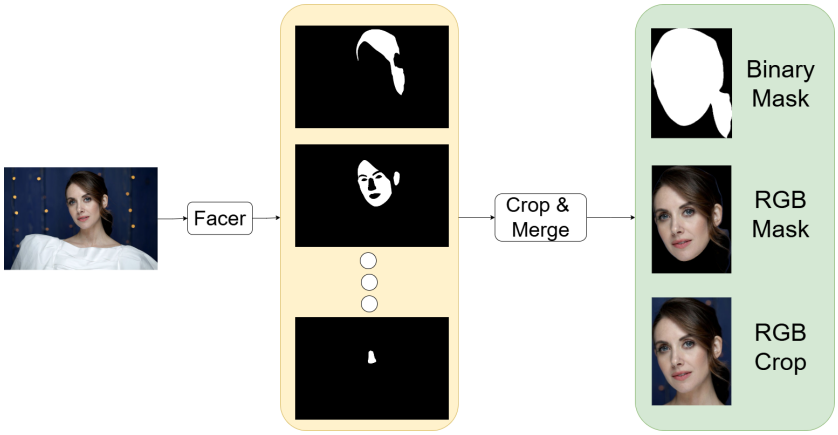


Fig. 1: Dataset processing with Facer to extract Face segmented masks.

Since the CelebA dataset already provides those data, we proceeded by isolating faces and their sub-parts within the Pivotal Armocromia set, exploiting the Facer toolbox, which is a DL-based Face Detection and Segmentation Module [7]. As visually reported in Figure 1, we exploited Facer to segment masks including pixels related to hair, face skin, nose, brows, eyes, and mouth. We then merge those segmentation masks to generate a unique mask that isolates only the pixels enclosing the features of interest. Finally, we visually process all the extracted face crops and relative segmentation masks to find wrong predictions. When finding a wrongly predicted mask/bounding box, we manually fixed that using the Gimp Image Editing software [8].

3.4 Dataset Statistics

In this Section, we report statistics obtained from the Deep Armocromia dataset and qualitative samples to visualize and understand how the single Armocromia classes were annotated. As visually reported in Figure 2, our dataset is composed of 4920 samples, categorized into four primary season classes: “Autumn”, “Summer”, “Winter”, and “Spring”.

[7] `https://github.com/FacePerceiver/facer`
[8] `https://www.gimp.org/`

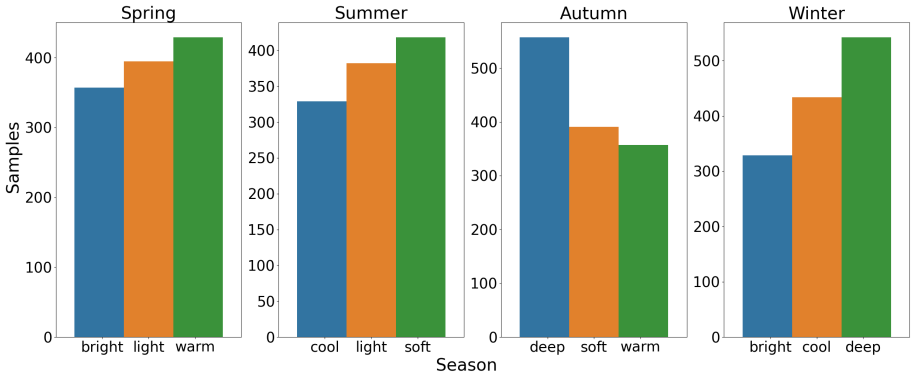


Fig. 2: Dataset statistics aggregated by main Armocromia Season Class and their Sub-Types.

Each season class is represented by a specific number of samples: "Autumn" and "Winter" both contain 1305 samples, "Spring" comprises 1181 samples, and "Summer" includes 1129 samples. These season classes are further subdivided into twelve distinct subtypes, reflecting more granular characteristics. The distribution of these subtypes is as follows: "Autumn Deep" has 557 samples, "Winter Deep" has 542 samples, "Winter Cool" has 434 samples, "Spring Warm" has 429 samples, "Summer Soft" has 418 samples, "Spring Light" has 395 samples, "Autumn Soft" has 391 samples, "Summer Light" has 382 samples, "Autumn Warm" and "Spring bright" each have 357 samples, and "Summer Cool" and "Winter bright" each have 329 samples. Additionally, we here report that Deep Armocromia samples are shared among 2981 samples from the CelebA dataset and 1939 samples from the Pivotal Armocromia Set, providing a comprehensive and diverse representation of seasonal and sub-type variations.

Figure 3 reports qualitative examples taken from Deep Armocromia, according to both Season classes and relative Sub-Types, sampled from both the Pivotal Set and the CelebA one.

It can be appreciated how, subjects from the Spring class are characterized by warm undertones in their skin and hair, while Summer ones exhibit cool, muted features with soft hair colors and eyes complemented by subtle and gentle color palettes. Autumn individuals have rich, warm tones with rich hair and eye colors, saturated colors that highlight their natural depth and warmth while Winter ones embody cool and intense characteristics, with variations including bright and icy features, but also high contrast and clear traits.

3.5 Benchmarks

We build two benchmarks by using the images and labels from Deep Armocromia. For each benchmark, there are 4k images for training and approx 900 images for testing (80% and 20% respectively) providing approximately the same balancing among all Season and Sub-Types.

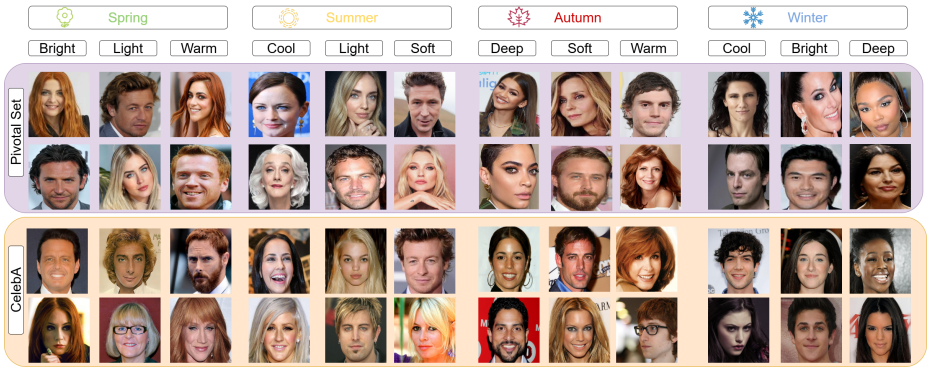


Fig. 3: Dataset examples coming from different Armocromia Seasons and Sub-Types.The main classes are Spring, Summer, Autumn, and Winter and their possible Sub-Types, which amount to Bright, Light, Warm, Cool, Soft, and Deep.

Season Classification This task aims at classifying the 4 coarse Armocromia classes. For category classification, we employ the standard accuracy, precision, recall, f1, and top-k classification accuracy as evaluation metrics.

Season Sub-Type Classification This task aims to classify the 12 fine Armocromia season sub-types. Also in this case we employed the standard accuracy, precision, recall, f1, and top-k classification accuracy as evaluation metrics.

4 Experimental Setting

In our experimental setting, we followed the rationale detailed in Figure 4 which will be detailed in the following.

Dataset Considering that our pipeline aims at classifying Armocromia Season and their Sub-Types by analyzing specific facial cues (i.e., hair, skin, and eyes), we first isolate them with the segmentation mask generated during the dataset collection. This defines RGB-masked images that are then used to optimize different baselines on the detailed benchmarks. In the process, we ensure that no face images overlap between training and testing sets.

Adopted Methods We considered a classical fine-tuning approach where we extract features from a pre-trained visual backbone (with frozen weights), that are forwarded to a trainable classifier. In our case, we considered two kinds of visual backbones. The first one amounts to a pre-trained Vision Transformer for Facial Representation Learning [31], where we considered two different backbones, trained respectively for 16 and 64 epochs (respectively (FaRL16 and FaRL64) on the LAION-Face dataset [31]. The second one amounts to pre-trained ResNeXt50 [29] on the ImageNet dataset.

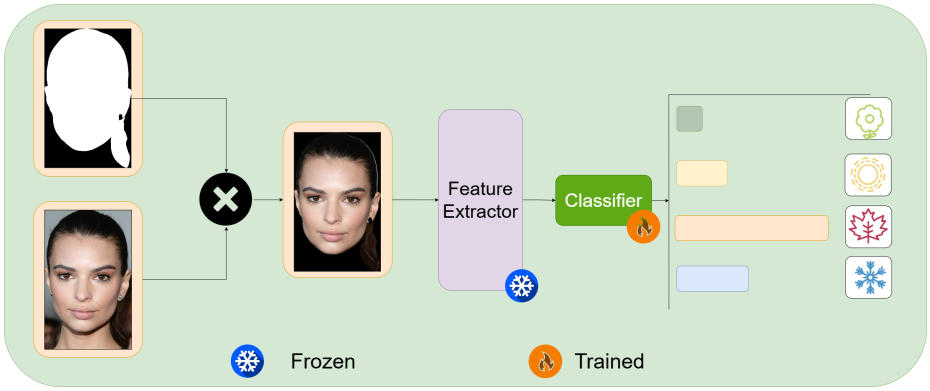


Fig. 4: Our system architecture is composed of a **Face Parsing** module which allows to select and isolate pixel areas for color analysis, which are then forwarded to a Deep Learning based **Classifier** which provides the most probable season class, employing pre-trained **Feature Extractors**.

This choice was driven by the fact that we wanted to evaluate whether Armocromia analysis must be performed exploiting specific learned facial features or could be achieved also considering more general ones [24].

The classifier, instead, is the same for each fine-tuning performed on those different backbones: it consists of two stacked Fully Connected (FC) layers, where the first reduces the input dimension (taken from the feature extractor) by half. The output of this layer is fed through the ReLU activation function and then to a dropout layer ($p = 0.5$) to reduce overfitting. The resulting vector is then fed to the second FC layer, which maps the reduced features to the number of target classes (4 and 12 respectively for the Season and Season Sub-Type classifications).

We outline that all the adopted methods were developed with Pytorch (v. 2.3.0), with CUDA 12.1 support for accelerated GPU computations. The pre-trained vision backbones were initialized through the Pytorch vision extensions (v. 0.18.0) and the Pytorch image models package [9] (v. 0.8.3). All the model's performance evaluations were measured through TorchMetrics [10] (v. 1.4.0).

Training Setting All the experiments were conducted on an Asus Rog Strix G814ji equipped with an Intel Core I9 13980 HX and an NVIDIA GeForce RTX 4070 with 8 GB of VRAM. To have a fair and meaningful comparison between the two considered backbones, the same training setting has been applied. The image dataset was resized to $3 \times 224 \times 224$ pixels as required by the selected backbones input layers, applying the following data augmentation to reduce overfitting: random crop, horizontal flipping with 50% probability, color jittering (brightness max factor=0.4, contrast max factor=0.2, saturation max

[9] `https://github.com/huggingface/pytorch-image-models`
[10] `https://github.com/Lightning-AI/torchmetrics`

factor=0.2), and random sharpness adjustments (sharpness factor = 2, p = 0.2) probability. We optimized all our models for 50 epochs using the AdamW optimizer (lr = 1e-3, weight decay = 1e-5) [13], using a Cosine Annealing scheduler with warm restarts (every 10 iterations, min lr = 1e-5) [12] and a batch size of 64.

5 Results

This section presents quantitative evaluations of various methods on the two defined benchmarks. We also explore the influence of the selected backbones in the final performance.

5.1 Season Classification

As reported in Table 2 the evaluation of model performances on the Season classification task outlined the FaRL64 as the best one among the three tested backbones.

Name Exp	Accuracy	Precision	Recall	F1 Score	Top-2
FaRL16	0.525	0.518	0.525	0.516	0.815
FaRL64	**0.554**	**0.553**	**0.554**	**0.548**	**0.808**
ResNeXt50	0.513	0.516	0.513	0.502	0.789

Table 2: Evaluation metrics for different model backbones on the Season classification task.

It achieves the highest accuracy of 0.554, precision of 0.553, recall of 0.554, and F1 score of 0.548, outperforming both FaRL16 and ResNeXt50. This performance can be attributed to its pre-training on the multi-modal LION-Face dataset, which allowed it to learn more complex feature extraction capabilities compared to ResNeXt50. Additionally, the top-2 accuracy is equal to 0.808 further reinforcing FaRL64's capability in providing reliable predictions. At the same time this shows that even if providing uncertainty in the final prediction, the Season Armocromia classification task amounts to an worth to explore classification task. Considering the FaRL64 the best model, we here provide its relative classification confusion matrix in Figure 5.

Notably, the model exhibits strong performance in predicting the Winter class, with 193 correct predictions, suggesting high precision and recall for this season. Conversely, there is significant confusion between Autumn and Winter, as evidenced by 80 instances of Autumn being misclassified as Winter. Additionally, the model struggles to distinctly classify Summer and Spring, with noticeable misclassifications across these classes (Summer misclassified as Spring in 59

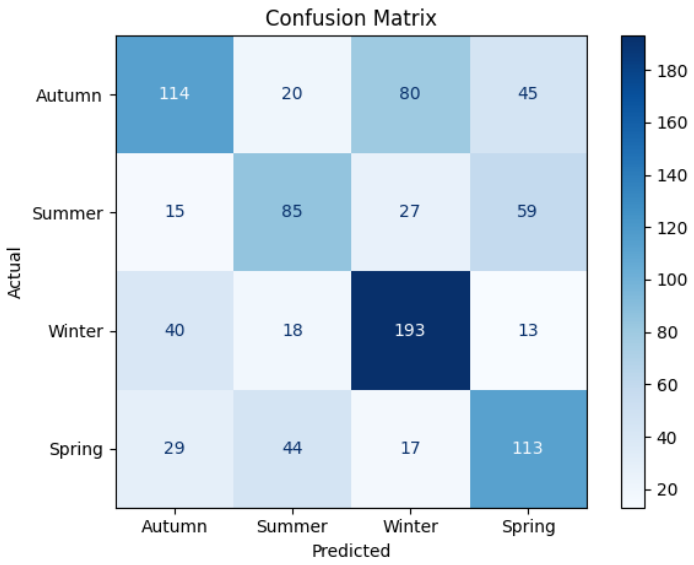


Fig. 5: Confusion Matrix of the best model, which amounts to FaRL64 fine-tuned for 50 epochs on the main Season classification task.

instances and Spring misclassified as Summer in 44 instances). These patterns may imply overlapping features between these two seasons in the dataset, which are indeed neighbors in the Armocromia classes dictionary and at the same time highlight the challenge behind Armocromia classification.

5.2 Season Sub-Types Classification

Table 3 presents the comprehensive evaluation metrics for various model backbones on the challenging task of Armocromatic Sub-Type classification across 12 distinct classes. In this case, we selected a Top-3 accuracy considering that we have 12 classes instead of 4.

Name Exp	Accuracy	Precision	Recall	F1 Score	Top-3
FaRL16	**0.318**	**0.317**	**0.318**	**0.310**	**0.663**
FaRL64	0.313	0.307	0.313	0.308	0.651
ResNeXt50	0.281	0.262	0.281	0.250	0.614

Table 3: Evaluation metrics for different model backbones on the Season and Sub-Types classification task.

Among the models assessed, FaRL16 achieves the highest overall performance metrics, with an accuracy of 0.318, precision of 0.317, recall of 0.318, and F1 score of 0.310. It also demonstrates the highest top-3 accuracy (0.663). In contrast, FaRL64 and ResNeXt50 exhibit slightly lower performance across all metrics, with FaRL64 achieving competitive results indicating that the features extracted by such backbones are semantically better than the ones of a classical backbone. The fact that the FaRL64 backbone performed worse than its FaRL16 counterpart for the Season Sub-Types classification may be attributed to the fact that it exploits broad features necessary for Season classification (coarse), at the expense of the subtle, detailed features required for fine-grained Armocromia classes. Here considering FaRL16 as the best model, we report in Figure 6 its classification confusion matrix.

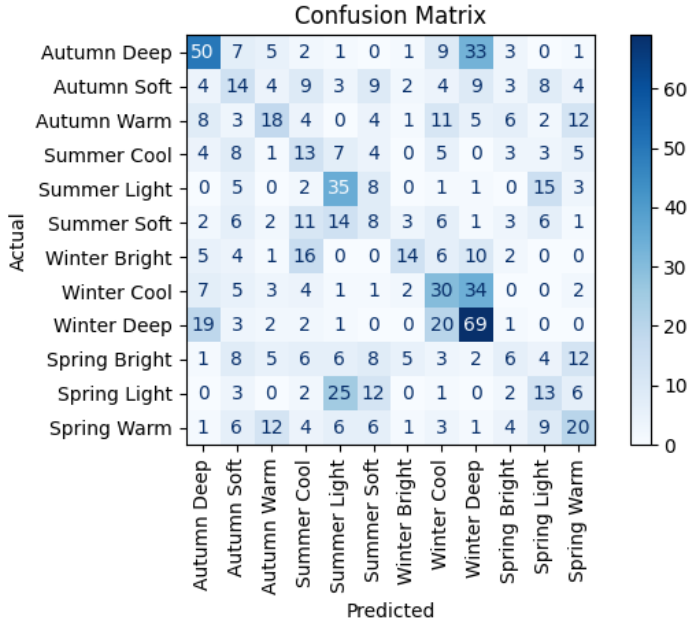


Fig. 6: Confusion Matrix of the best model, which amounts to FaRL16 fine-tuned for 50 epochs on the Season and Sub-Type classification task.

The confusion matrix indicates notable misclassifications between certain Armocromia Sub-Types. Specifically, there is significant confusion between “Autumn Deep" and “Winter Deep" with 19 samples of “Autumn Deep" being misclassified as “Winter Deep" This misclassification likely arises from the similar characteristics shared by these categories, such as the depth and intensity of colors. Furthermore, the “Summer Cool" and “Winter Cool" classes exhibit a high level of mutual confusion, with 13 samples of “Summer Cool" being classified

as “Winter Cool" and 12 samples of “Winter Cool" being classified as “Summer Cool". This confusion could be attributed to the overlapping cool tones present in both categories. Additionally, the “Spring Bright" class shows significant confusion with “Spring Light" and “Spring Warm" indicating that the distinguishing features between the spring subcategories are not well captured by the model. To conclude, the performance obtained in the Season Sub-Type classification is overall lower than the ones reported for the Season Classification benchmark, indicating a higher degree of challenge towards the distinctions of those 12 Armocromia classes.

6 Conclusion

In this work, we have addressed the challenge of applying DL approaches to the domain of seasonal color analysis of Faces (Armocromia). We introduced the Deep Armocromia dataset, a novel and comprehensive collection of 5000 face images of celebrities, labeled according to the strict protocols of Armocromia and Flow Theory, including the 4 Season classes and their Sub-Types. We also carried out an experimental campaign that demonstrated the challenging level of the Armocromia classification task, in particular for the Season Sub-Types. Future work will focus on augmenting the dataset with more samples for training highly robust deep learning models, particularly for more granular subtype classifications while exploring novel DL techniques to improve classification accuracy. In particular, we will explore those paradigms involving hierarchical and ordinal learning, given that such nature matches the inner nature of the Armocromia classes [21]. We will also implement a systematic approach to verify whether DL paradigms provide an advantage over classical computer vision and image processing pipelines for Armocromia classification.

Acknowledgments

This work is funded by Moda Metrics s.r.l, which also supported us in the Deep Armocromia data collection process.

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